

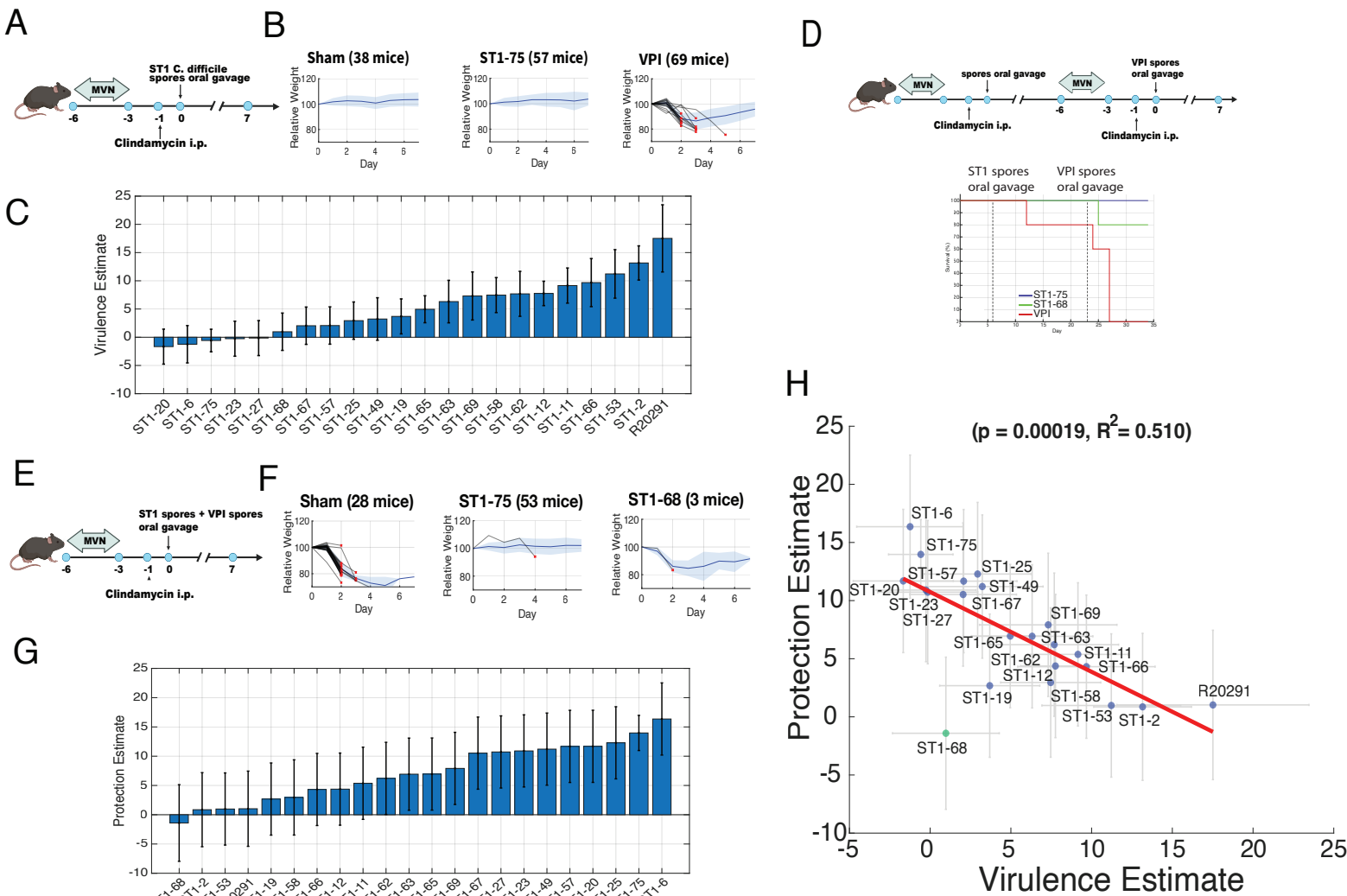


Figure 1: Clinical ST1 isolates of *C. difficile* vary in virulence and protection against VPI10463 challenge. (A) Schematic of the mouse model used to assess virulence. (B) Representative relative weight trajectories during primary infection for sham (PBS, n=38), ST1.75 (n=57) and VPI10463 (n=69) infected mice. Shaded regions indicate 95% confidence interval. (C) Virulence scores for all 21 ST1 clinical isolates along with virulent R20291 strain, derived from linear mixed-effects models of relative weight loss trajectories during primary infection. The model used sham-infected mice as the reference, such that positive effect sizes reflect greater weight loss (higher virulence). Error bars indicate standard error. (D) Survival curves during the secondary VPI10463 rechallenge for mice precolonized with ST1.75 (blue), ST1-68 (green), or VPI10463 alone (red) (5 mice in each group). The first infection was administered on day 6 and the VPI10463 rechallenge on day 23 (dashed vertical lines). (E) Schematic of the co-infection mouse model used to assess protection. Mice were given with the same antibiotic regimen as shown before and then co-infected with VPI10463 spores together with spores of an individual ST1 strain via oral gavage. (F) Representative relative weight trajectories during co-infection for sham+VPI (PBS; n=28), ST1.75+VPI (n=53), and ST1.68+VPI (n=3) groups. Shaded regions indicate 95% confidence intervals. (G) Protection scores for all 21 ST1 clinical isolates, derived from linear mixed-effects models of relative weight loss during VPI10463 co-infection. The model used sham+VPI mice as the reference, such that positive effect sizes reflect weight preservation (greater protection). Error bars indicate standard error. (H) Relationship between virulence score (x-axis) versus protection score (y-axis) for all 21 ST1 isolates. A significant negative linear relationship was observed ($R^2 = 0.510$, $p = 0.00019$) shown here in red line, indicating that higher virulence correlates with reduced protection against VPI10463. The outlier ST1.68 strain, colored in green, disagreed with that trend (statistically significant outlier by Bonferroni-corrected externally studentized residual test ($t = -5.16$, $p = 0.0012$))